Jiin Choi

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RESEARCH INTEREST

My work lies at the intersection of human-computer interaction (HCI) and artificial intelligence (AI), focusing on interactive intent alignment for design-space exploration. Specifically, I study how to decompose user prompts, interaction logs, and model responses into design elements that enable users to (1) iteratively refine their intents to guide GenAI-driven design tools, and (2) audit and steer the model's behavior to ensure it aligns with those intents and supports better user experience in individual cognitive space.

EDUCATION

Mar 2023 - Present
Ph.D. & M.S. Integrated Degree
Interior Architecture Design · Hanyang University, Seoul
Advised by Prof. Kyung Hoon Hyun
Design AI Lab & Human-Centered AI Design Institute

Mar 2018 - Feb 2023
Bachelor of Science in Interior Architecture Design
Minor in Computer Science
Hanyang University, Seoul

PUBLICATIONS

† Equal contribution · CA Corresponding author

Conference and Journal Papers - International

[C3] CogChat: Knowledge Graph-Augmented Conversational AI with Heterogeneous Graph Transformer for Cognitive Grounding in Design Generation
Choi, J. & Hyun, K. H.[CA] (2026).
Proc. ACM Symposium on User Interface Software and Technology(UIST '26)
BK TOP-TIER · H5 53 · FIRST **DOI**

[J6] How Agent-Driven Creative Trace Mining Reshapes Design Behavior in a Deployment Study of Intent-Aligned Multimodal Design
Lee, S. W.†, **Choi, J.**†, Yun, Y.†, Jin, S., Jang, Y., Hwang, G., Park, S. W., Ban, S., & Hyun, K. H.[CA] (2026).
International Journal of Human-Computer Interaction
SSCI · Top 8% (Ergonomics, IF 6.100) · CO-FIRST **DOI**

[J5] GenAlign: Visual Knowledge Graphs as Interactive Substrates for Property-Aligned Generative Design
Choi, J., Seo, M., & Hyun, K. H.[CA] (2026).
Advanced Engineering Informatics (Accepted)
SCIE · Top 2% · IF 11.500 · FIRST **DOI**

[J4] Workflow-Net: Toward Understanding Designer Workflows in Generative AI-Driven Systems through Comparing Node- and Prompt-Based Interfaces
Jo, T. H.†, **Choi, J.**†, Jin, S., Lee, S. W., Jang, Y., Park, S. W., Ban, S., & Hyun, K. H.[CA] (2026).
Archives of Design Research, vol. 39, no. 2 (2026): 7-34
SCOPUS · CO-FIRST · Cover Paper · The Best Research Paper Award **DOI**

[J3] Part-level 3D Shape Generation Driven by User Intention Inference with Preferential Bayesian Optimization
Lee, S. W., **Choi, J.**, & Hyun, K. H.[CA] (2026).
Scientific Reports, vol. 16, Article 7715
SCIE · IF 4.6 DOI

[C2] GenPara: Enhancing the 3D Design Editing Process by Inferring Users' Regions of Interest with Text-Conditional Shape Parameters
Choi, J., Lee, S. W., & Hyun, K. H.[CA] (2025).
Proc. ACM CHI Conference on Human Factors in Computing Systems (CHI '25), Yokohama, Japan
BK TOP-TIER · H5 139 · FIRST DOI

[J2] CMLsearch: Semantic Visual Search and Simulation through Segmented Colour, Material, and Lighting in Interior Image
Jin, S.†, **Choi, J.**†, & Hyun, K. H.[CA] (2025).
Journal of Computational Design and Engineering, 12(1), 179-299
SCIE · CO-FIRST DOI

[C1] The Impact of Sketch-guided vs. Prompt-guided 3D Generative AIs on the Design Exploration Process
Lee, S. W., Jo, T. H., Jin, S., **Choi, J.**, Yun, K., Bromberg, S., Ban, S., & Hyun, K. H.[CA] (2024).
Proc. ACM CHI Conference on Human Factors in Computing Systems (CHI '24), Honolulu, HI, USA
BK TOP-TIER · H5 139 DOI

[J1] Typeface Network and the Principle of Font Pairing
Choi, J., & Hyun, K. H.[CA] (2024).
Scientific Reports, vol. 14, Article 30820
SCIE · FIRST DOI

Posters, Demos, and Workshop Papers

[P1] Modeling Sequential Design Actions as Designer Externalization on an Infinite Canvas
Yun, Y.†, Lee, S. W.†, **Choi, J.**†, & Hyun, K. H.[CA] (2026).
CHI '26 Extended Abstracts, Barcelona, Spain
CO-FIRST DOI

[W1] Toward AI-driven Multimodal Interfaces for Industrial CAD Modeling
Choi, J.†, Jang, Y.†, & Hyun, K. H.[CA] (2025).
CHI '25 Workshop · Bridging HCI and Industrial Manufacturing
CO-FIRST PDF

Conference and Journal Papers - Domestic / 국내

[DC8] A Multi-Agent Guidance System with Evidence Confidence for Reviewing Spatial Analysis Checklists
다중 에이전트와 증거 신뢰도 기반의 공간 분석 체크리스트 가이드 시스템
최지인 , 장유경, 현경훈[CA]
2026 한국실내디자인학회 학술발표대회 논문집, 28(1), 140-143

[DC7] A Part-based Workflow Visualization Interface for 3D Furniture Design from Multimodal Inputs
파트 기반 워크플로우 시각화를 위한 3D 가구 디자인 인터페이스 제안
장유경, **최지인** , 현경훈[CA]
2026 한국실내디자인학회 학술발표대회 논문집, 28(1), 105-107

[DC6] An Agent-based Guidance System for Spatial Analysis Checklist Review
Choi, J., & Hwang, Y. S. (2025). 공간 분석 체크리스트 점검을 위한 에이전트 기반 가이드 시스템.
Proceedings of KSDS Annual Conference, 40-41
FIRST

[DC5] An LLM-driven Font Pairing Guide System Based on Design Intent
Choi, J., & Hyun, K. H.[CA] (2025). 대규모 언어 모델을 활용한 디자인 의도 기반 폰트 조합 가이드 시스템.
Proceedings of KSDS Annual Conference, 48-49
FIRST

[DC4] An LLM-driven Method for Element-level 3D Shape Editing
Choi, J., Lee, S. W., Jang, Y., Jin, S., Jo, T. H., & Hyun, K. H.[CA] (2024). LLM을 활용한 디자인 요소 단위 3차원 형태 편집 방법 제안.
Proceedings of KSDS Annual Conference, 108-109
FIRST

[DC3] An Element-level 3D Shape Editing Method via Latent-Space Reduction in 3D Generative AI
Lee, S. W., **Choi, J.**, Jin, S., Jo, T. H., Jang, Y., & Hyun, K. H.[CA] (2024). 3차원 생성형 AI의 잠재 공간 축소를 통한 디자인 요소 단위 3차원 형태 편집 방법 제안.
Proceedings of KSDS Annual Conference, 56-57

[DC2] A 3D Scene-Graph-Based Interface for Spatial Generation and Editing using Colour & Material Information
Jin, S., Jang, Y., **Choi, J.**, Jo, T. H., Lee, S. W., & Hyun, K. H.[CA] (2024). 색채, 재질 정보 중심의 3차원 장면 그래프 기반 공간 생성 및 편집 인터페이스 제안.
Proceedings of KSDS Annual Conference, 236-237

[DC1] Multi-View Image Generation with a Single Sketch Image
Lee, S. W., **Choi, J.**, & Hyun, K. H.[CA] (2024).
KSDS International Invited Exhibition Catalogue, 123

Under Review

[U1] Taxonomy X Interaction
Choi, J., ..., & Hyun, K. H.[CA]
~SCIE · FIRST AUTHOR

[U2] Ambiguity based on Knowledge-Graph for Authoring
Choi, J., & Hyun, K. H.[CA]
~SCIE · FIRST AUTHOR

[U3] Recontextualization
Choi, J.†, ...†, & Hyun, K. H.[CA]
SSCI · CO-FIRST AUTHORS

[U4] Universal 3D Furniture Design
..., ..., ..., **Choi, J.**, & Hyun, K. H.[CA]
~SCIE

[U5] Preference-based design trade-off management in 3D generation
..., **Choi, J.**, & Hyun, K. H.[CA]
SCIE

PATENTS

14 filings / registrations (including 2 design registrations)

System and Method for Personalized Design Generation Based on a Visual Attribute Knowledge Graph
시각 속성 지식그래프 기반 개인화 디자인 생성 시스템 및 그 방법
Hyun, K. H., & **Choi, J.**
REGISTERED KR 10-3019833 (App. 10-2025-0215047) · Registered Sep 10, 2026

Personal Knowledge Graph-Augmented Conversational Design System Based on Heterogeneous Graph Transformer
이질적 그래프 트랜스포머 기반 개인 지식그래프 대화형 디자인 생성 시스템 및 그 방법
Hyun, K. H., & **Choi, J.**
FILED KR 10-2026-0170596 · Filed Sep 8, 2026

Display Panel with a Graphical User Interface
화면디자인이 표시된 디스플레이 패널
REGISTERED · DESIGN KR 30-1348145 (App. 30-2026-0004674) · Filed Feb 3, 2026 · Registered Apr 23, 2026

Display Panel with a Graphical User Interface
화면디자인이 표시된 디스플레이 패널
REGISTERED · DESIGN KR 30-1348142 (App. 30-2026-0004671) · Filed Feb 3, 2026 · Registered Apr 23, 2026

User Intent-Based Context-Reflective Product Image Generation System and Method
사용자 의도 기반 맥락 반영형 제품 이미지 생성 시스템 및 방법
Hyun, K. H., **Choi, J.**, & Jang, Y.
REGISTERED KR 10-2957285 (App. 10-2026-0033081) · Registered Apr 21, 2026

Method and Apparatus for Providing Interior Product Search Service
인테리어 제품 검색 서비스 제공 방법 및 장치
Hyun, K. H., Jin, S., & **Choi, J.**
REGISTERED KR 10-2943193 (App. 10-2023-0171111) · Registered Mar 19, 2026

System and Method for Design Artifact-Based Image Generation
디자인 아티팩트 기반 이미지 생성 시스템 및 방법
Hyun, K. H., Lee, S. W., **Choi, J.**, & Jin, S.
FILED KR 10-2025-0184966 · Filed Nov 28, 2025

System and Method for Analyzing Action-Centered Design Process Workflows
행동 중심 디자인 과정 워크플로우 분석 시스템 및 방법
Hyun, K. H., Jo, T. H., **Choi, J.**, & Lee, S. W.
FILED KR 10-2025-0184967 · Filed Nov 28, 2025

Method and System for Generating Advertising Content
광고 콘텐츠 생성 방법 및 시스템
Hyun, K. H., Lee, S. W., **Choi, J.**, & Jang, Y.
FILED KR 10-2025-0070339 · Filed May 29, 2025

Text-Conditional Shape Parameter Utilization System for Inferring Design Preferences in 3D Design Space
3D 디자인 공간 속 디자인 선호 추론을 위한 텍스트 조건부 형상 매개변수 활용 시스템
Hyun, K. H., & **Choi, J.**
FILED KR 10-2024-0054885 · Filed Apr 24, 2024

Method of Generating Design Based on User Preference
사용자 선호도 기반의 디자인 생성 방법
Hyun, K. H., Lee, S. W., & **Choi, J.**
REGISTERED KR 10-2023-0159030 / 10-2792802 · Registered Apr 3, 2024

3D Shape Manipulation Method and System with Multi-objective Preferential Bayesian Optimization
다목적 선호도 베이지안 최적화 기반 3D 형상 편집 시스템 및 그 방법
Hyun, K. H., Lee, S. W., & **Choi, J.**
FILED KR 10-2024-0054886 · Filed Apr 22, 2024

3D Design Trade-off Optimization Support System
3D 형태 편집에서 다목적 선호도 베이지안 최적화를 통한 디자인 선호도 트레이드오프의 균형을 지원하는 시스템 및 그 방법
Hyun, K. H., Lee, S. W., & **Choi, J.**
FILED KR 10-2025-0053863 · Filed Apr 24, 2024

Method of Generating Design Based on User Preference
- US continuation
Hyun, K. H., Lee, S. W., & **Choi, J.**
FILED US 18/411,182 · Filed Jan 12, 2024

RESEARCH EXPERIENCE

Oct 2022 - Feb 2023

Research Intern
Design Informatics Lab (DILAB), Hanyang University
Collected and cleaned project data (images, logs, surveys), structured it into a schema/DB, and conducted statistical and qualitative analyses to produce actionable reports.

TEACHING

Fall 2026

Teaching Fellow (4 weeks) · AIxDesignResearch (HOM4041)
Department of Interior Architecture Design, Hanyang University
Instructor: Prof. Kyung Hoon Hyun

Spring 2026

Teaching Fellow (4 weeks) · Design Computing (HOM4025)
Department of Interior Architecture Design, Hanyang University
Instructor: Prof. Kyung Hoon Hyun · English-taught course

2023-2025

Teaching Assistant · Design Computing (HOM4025)
Department of Interior Architecture Design, Hanyang University
Instructor: Prof. Kyung Hoon Hyun · English-taught course

ACADEMIC SERVICE

Reviewer

2026

Conference reviewing
ACM UIST 2026 - Paper
ACM CHI 2026 - Poster
ACM DIS 2026 - Paper · Poster
ACM C&C 2026
IMX 2026 - Technical Paper
ACM IUI 2026 - Poster

2025

Conference reviewing
ACM UIST 2025 - Paper
ACM CHI 2025 - Paper

Workshops & Talks Attended

CHI 2026

Sensemaking Workshop
Exploring how designers make sense of AI outputs across iterations.

UIST 2026

Mind & Motion Workshop
Exploring how biomechanical and cognitive models can inform the design of interactive systems.

CHI 2025

Bridging HCI and Industrial Manufacturing Workshop
Exploring how HCI research and industry collaboration can support human-centered manufacturing.

INDUSTRY

Jul 2019 - Aug 2022

Design Intern
One Day One Communication (ODOC), Seoul
Ran user research to capture requirements, produced wireframes, mockups, and prototypes, and structured and analyzed usability findings to deliver design recommendations.

Jul 2020 - Dec 2020

Architecture / Interior Design Intern
Arch Architects, Seoul
Produced CD sets and detail drawings, organized spec/schedule sheets, and coordinated code/consultant updates for build-ready documentation.

AWARDS

Global Design iT Award
GenPresso

RED Award
제24회 대한민국색채대상
GenPresso

NY Design Award
GenPresso
Award details

Feb 2023

Stockholm Furniture Fair Exhibition
WAS (Waste As Source) Project · Stockholm, Sweden

Nov 2022

Honorable Mention
The 35th Space Prize for International Students of Architecture Design
Work: Relocated Cinema

2021

Excellent Work
The Klasse 2021 - Indoor Architecture Competition
Work: Société #

2020

Excellent Work
Busan Citizen Idea Contest - Wanwol-dong Alley Regeneration